Year 8
Mathematics

Unit 4: 2D Shapes and 3D Solids

4.4 Surface Area

Surface Area

STARTER · 5 MIN

Date: _____ **Title: Surface Area**

LEARNING OBJECTIVES

- Calculate the surface area of cubes and cuboids

STARTER *Answer in your book. No calculators.*

1. Work out 3 × 3
2. Work out 6 × 9
3. Area of a rectangle 6 cm by 4 cm
4. Work out 24 + 12 + 8

Starter — Answers

STARTER

Tick or correct in green pen

1. **9**

2. **54**

3. **24 cm²**

4. **44**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can find the area of squares and rectangles	☐
SILVER	I can solve real-life surface area problems	☐
GOLD	I can use the surface area formula for cubes and cuboids	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Surface Area

INTRODUCTION · 3 MIN

Surface area is the total area of all the faces of a 3D solid.

Cube

6 equal square faces.

SA = 6 × side × side

Cuboid

3 pairs of equal faces.

SA = 2(*lw* + *lh* + *wh*)

Units

Area is in square units.

cm^2, m^2

Net

Draw the net to see every face.

A cuboid net has 6 rectangles

Examples 1 and 2 — Cubes

EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

Hint Cube: 6 × side × side Units: cm^2

Example 1

Find the surface area of a cube of side 3 cm.

Example 2

Find the surface area of a cube of side 5 cm.

Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

6 equal squares

one face $= 3 \times 3 = 9\ cm^2$

SA $= 6 \times 9$

$\mathbf{= 54\ cm^2}$

Tip: a cube has 6 faces.

Example 2

6 equal squares

one face $= 5 \times 5 = 25\ cm^2$

SA $= 6 \times 25$

$\mathbf{= 150\ cm^2}$

Tip: the units are cm^2.

Task 1

PRACTICE · 9 MIN

BRONZE

Find the surface area of a cube with side

Hint Cube: 6 × side × side Units: cm^2

1. 2 cm
2. 4 cm
3. 10 cm
4. 7 cm
5. 6 cm
★ 1.5 cm

Task 1 — Answers

CORRECTIONS

BRONZE

Tick or correct in green pen

1. **24 cm²**
2. **96 cm²**
3. **600 cm²**
4. **294 cm²**
5. **216 cm²**

★ **13.5 cm²**

Examples 3 and 4 — Cuboids

EXAMPLES · 5 MIN

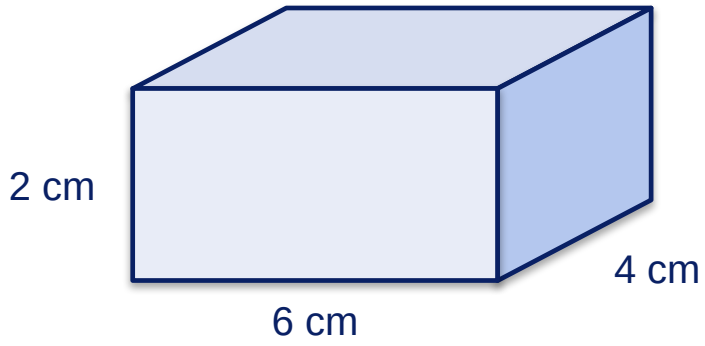
SILVER

Watch and copy into your book

Hint SA = 2(*lw* + *lh* + *wh*) Opposite faces are equal

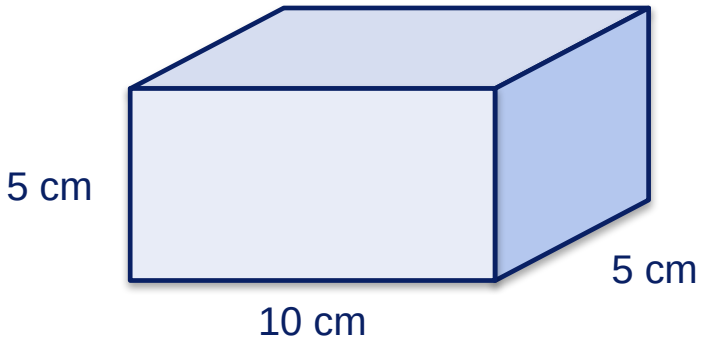
Example 3

Find the surface area.



Example 4

Find the surface area.



Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

3 pairs of faces

$$\begin{aligned} SA &= 2(6 \times 4 + 6 \times 2 + 4 \times 2) \\ &= 2(24 + 12 + 8) \\ &= \mathbf{88\ cm^2} \end{aligned}$$

Tip: front, top and side, then double.

Example 4

2 squares + 4 rectangles

$$\begin{aligned} SA &= 2 \times 25 + 4 \times 50 \\ &= 50 + 200 \\ &= \mathbf{250\ cm^2} \end{aligned}$$

Tip: the two ends are 5 by 5 squares.

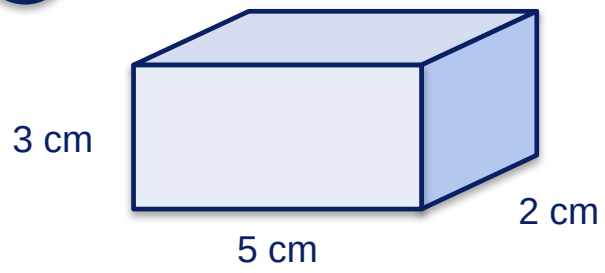
Task 2

PRACTICE · 9 MIN

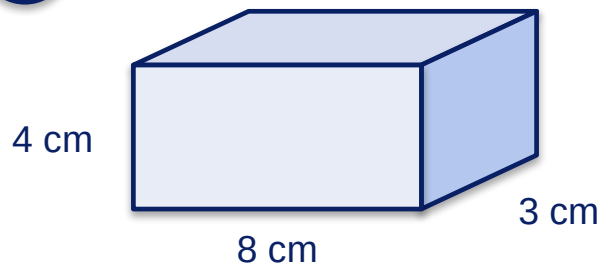
SILVER *Find the surface area*

Hint SA = 2(*lw* + *lh* + *wh*) Opposite faces are equal

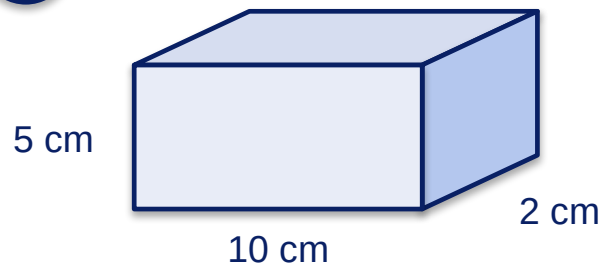
1 Surface area



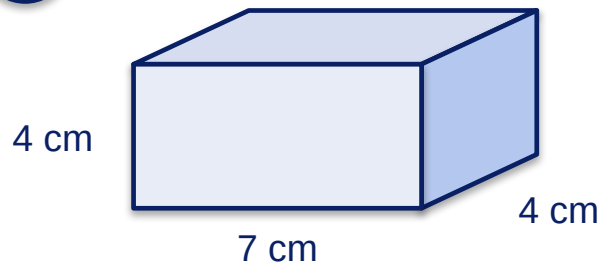
2 Surface area



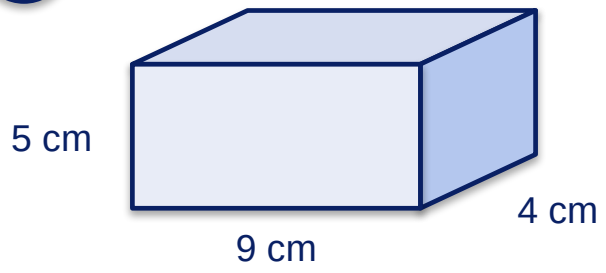
3 Surface area



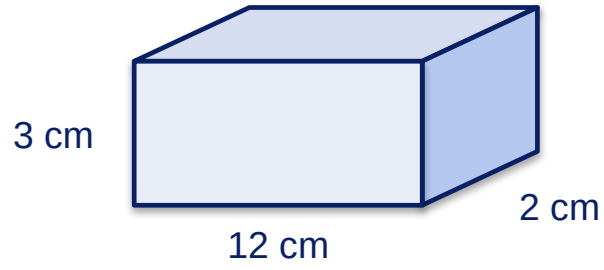
4 Surface area



5 Surface area



★ Surface area



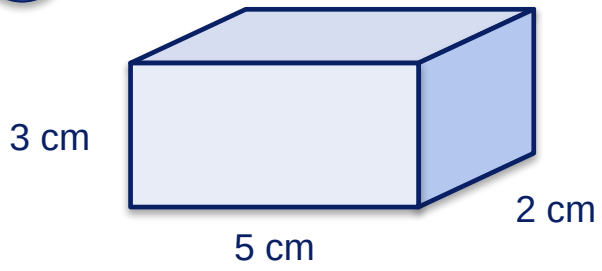
Task 2 — Answers

CORRECTIONS

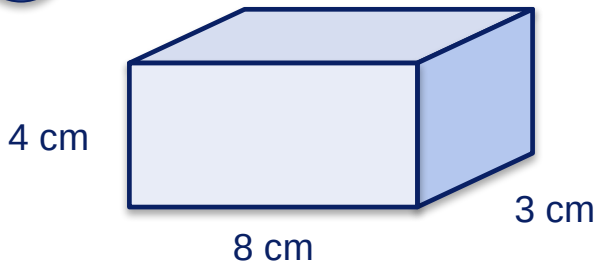
SILVER

Tick or correct in green pen

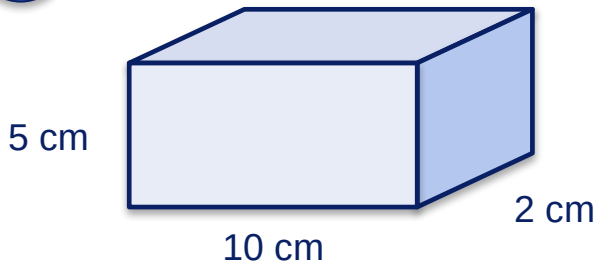
1 **62 cm²**



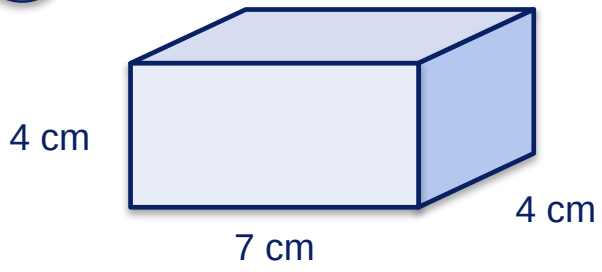
2 **136 cm²**



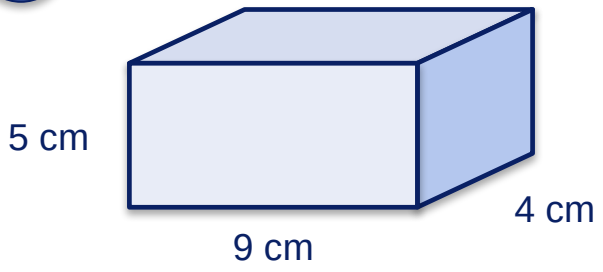
3 **160 cm²**



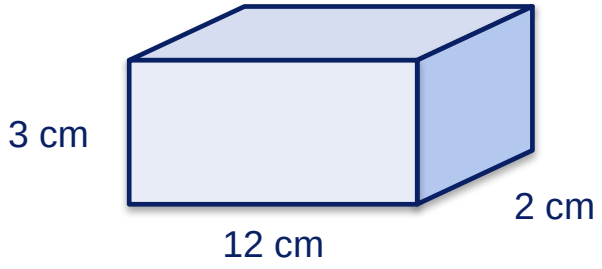
4 **144 cm²**



5 **202 cm²**



★ **132 cm²**



Examples 5 and 6 — Working Backwards

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint Cube: one face = SA ÷ 6 Then square root

Example 5 (2 marks)

The surface area of a cube is 96 cm^2.

Find the length of one side.

Example 6 (2 marks)

A cube has side n cm.

Write an expression for its surface area.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

Divide by 6, then square root

one face $= 96 \div 6 = 16\ \text{cm}^2$

side $= \sqrt{16}$

$= 4$ cm

Tip: $4 \times 4 = 16$.

Example 6

6 faces of $n \times n$

one face $= n^2$

SA $= 6n^2\ \text{cm}^2$

Tip: $n \times n = n^2$.

Task 3

PRACTICE · 9 MIN

GOLD *Exam-style: surface area problems*

Hint Cube: one face = SA ÷ 6 Then square root

1. Cube SA 150 cm^2. Side?

2. Cube SA 24 cm^2. Side?

3. Cube of side $2n$. SA?

4. Cuboid 8 cm by 5 cm by h cm, SA 184 cm^2. Find h.

5. Open box (no lid) 10 cm by 10 cm by 5 cm. SA?

★ Box 20 cm by 10 cm by 10 cm. Paper costs 1p per 10 cm^2. Cost?

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. **5 cm**
2. **2 cm**
3. **$24n^2$**
4. **$h = 4$ cm**
5. **300 cm²**

★ **1000 cm²: £1**

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Surface area of a cube of side 4 cm

SILVER Surface area of a cuboid 5 cm by 2 cm by 1 cm

GOLD A cube has SA 54 cm^2. Find its side.

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE **96 cm²**

SILVER **34 cm²**

GOLD **3 cm**

Exam Practice — Cubes and Cuboids

EXAM PRACTICE · 10 MIN

BRONZE

IGCSE exam-style: show all your working

Hint Cube: 6 × side × side Units: cm^2

Question 1

(2 marks)

A cube has edges of 6 cm.
Work out its surface area.

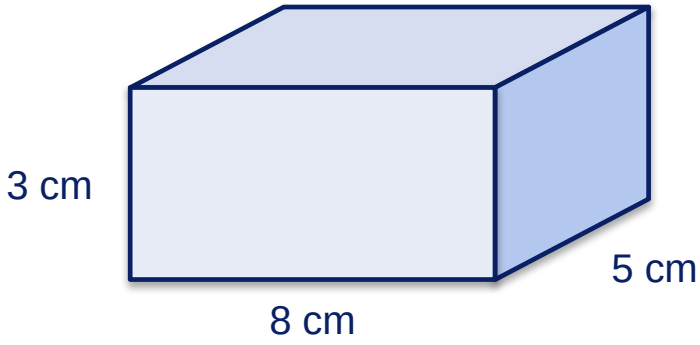
..... cm^2

Question 2

(3 marks)

Work out the surface area
of this cuboid.

Diagram NOT accurately drawn



..... cm^2

Exam Practice — Surface Area

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

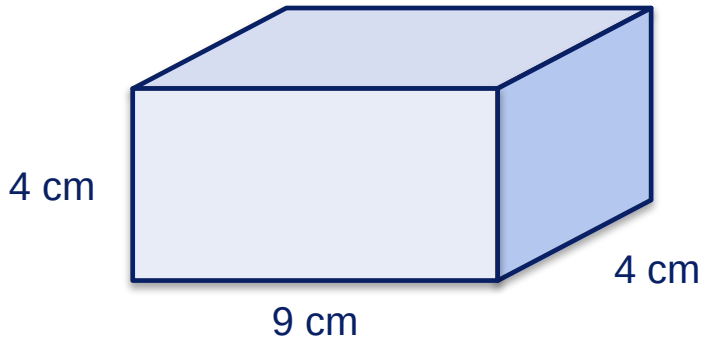
Hint
SA = 2(*lw* + *lh* + *wh*)
Opposite faces are equal

Question 3

(3 marks)

Work out the surface area of this cuboid.

Diagram NOT accurately drawn



..... cm^2

Question 4

(2 marks)

A cube has a surface area of 294 cm^2.

Work out the length of one side.

..... cm

Exam Practice — Surface Area Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint Cube: one face = SA ÷ 6 Then square root

Question 5

(4 marks)

A box 30 cm by 20 cm by 10 cm is painted all over. One tin covers 1500 cm^2.

How many tins are needed?

.....

Question 6

(3 marks)

A cuboid is 10 cm by 6 cm by h cm.
Its surface area is 248 cm^2.

Work out h.

..... cm

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

SA $= 6 \times 36$ **M1**

$\mathbf{= 216\ cm^2}$ **A1**

Tip: 6 faces of 36 cm^2.

Question 2

SA $= 2(8 \times 5 + 8 \times 3 + 5 \times 3)$ **M1**

$= 2(40 + 24 + 15)$ **M1**

$\mathbf{= 158\ cm^2}$ **A1**

Tip: 79 (only three faces) scores M1.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

SA $= 2 \times 16 + 4 \times 36$ — M1

$= 32 + 144$ — M1

$= 176\ cm^2$ — A1

Tip: two square ends, four rectangles.

Question 4

one face $= 294 \div 6 = 49$ — M1

side $= 7$ cm — A1

Tip: $7 \times 7 = 49$.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

SA	$= 2(600 + 300 + 200)$	M1
	$= 2200\ \text{cm}^2$	A1
tins	$= 2200 \div 1500 = 1.47$	M1
	= 2 tins	A1

Tip: round up the number of tins.

Question 6

$2(60 + 10h + 6h)$	$= 248$	M1
$60 + 16h$	$= 124$	M1
h	**= 4 cm**	A1

Tip: halve 248 first.